Pre-work · Joints

Articulations and Joints. Read the Part A chapter first. Every tagged joint can be asked four ways, so practise answering all four every time.

Module 2 · 4 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the joints chapter into mini visuals. Seven chunks. The two classification grids are the spine of the whole chapter.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

Classify the pubic symphysis by structure and by function, and name the tissue involved.

Answer. Structurally a cartilaginous joint, specifically a symphysis. Functionally an amphiarthrosis. The tissue is fibrocartilage.

Reasoning. Run the two classifications separately, then check they agree. Structure first: is there a synovial cavity? No, so it is fibrous or cartilaginous. What holds the bones? A broad flat disc of fibrocartilage, which is the definition of a symphysis. Function second: fibrocartilage allows a little give, so slightly movable, an amphiarthrosis. The two answers are consistent, which is the check that you classified it correctly.

Set A · Classify it

1. A tooth in its socket. Structural class ______________ Functional class ______________

takes longer than three minutes you are copying instead of organizing.

1. **GRID**

Structure crossed with function

Fibrous, cartilaginous, synovial down the side. Cavity present, tissue holding the bones, and usual functional class across the top. This one grid answers most exam questions in this chapter.

2. **HUB**

Fibrous joints

Fibrous at the hub. Spoke to suture, syndesmosis, gomphosis, interosseous membrane, synostosis. One example on each spoke.

3. **SPLIT**

Synchondrosis against symphysis

One split. Cartilage type, mobility, and two examples on each branch. The cartilage type is what decides everything else.

4. **HUB**

A synovial joint in section

Joint cavity at the hub. Spoke to articular cartilage, capsule with its two membranes, ligaments, synovial fluid, fat pads, menisci, labrum, bursae.

5. **GRID**

The six synovial types

Type down the side. Surface shape, axes, and one example across the top. Six rows, and the axes column is what people lose.

6. **GRID**

The movements

Two grids. Angular movements in one, special movements in the other. Draw a tiny arrow showing each direction rather than writing a definition.

DRAWING 1

A synovial joint, sectioned

BRAIN DUMP FIRST

Two minutes. Every structure inside a synovial joint and what each one does.

Draw two bone ends meeting, with a gap between them. Cap each with articular cartilage. Draw the capsule wrapping the joint, with the fibrous membrane outside and the synovial membrane inside, and fill the cavity with synovial fluid. Add a meniscus between the surfaces, a fat pad, and a bursa outside the capsule. Label all of it.

In your notebook, head the page **M2-1 A synovial joint, sectioned** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name the four jobs of synovial fluid while pointing at where it sits.

2. The elbow. Synovial type _______________ Axes _______________
3. The epiphyseal plate. Structural class _______________ Functional class _______________
4. The carpometacarpal joint of the thumb. Synovial type _______________ Axes _______________
5. The atlantoaxial joint. Synovial type _______________ Movement it produces _______________

Set B · Name the movement

1. Turning the sole of the foot inward is _________________________.
2. Moving the thumb across the palm to touch the fingertips is _________________________.
3. Turning the palm to face posteriorly is _________________________.
4. Bending the head sideways toward the shoulder is _________________________.
5. Name the movements possible at a ball-and-socket joint.

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Set C · Predict it

1. A lateral blow lands on a planted knee. Name the three structures likely torn and the name of the injury.

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2. A patient has an unstable shoulder after an injury. Explain why the ligaments alone do not stabilize it, and name what does.

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3. A knee swells immediately after an injury, and another swells the next day. Say what each timing suggests.

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7 **SPLIT**

Osteoarthritis against rheumatoid arthritis

One split. Cause, which joints, and whether it is symmetrical. Three branches each. This pair appears on every exam.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The knee, anterior and from above

BRAIN DUMP FIRST

Two minutes. Every ligament and both menisci.

Draw the knee from the front with the patella lifted away: femoral condyles, tibial plateau, both collateral ligaments outside, both cruciates crossing inside. Then draw the tibial plateau from above and put the C-shaped medial meniscus and the O-shaped lateral meniscus in place. Mark the three structures of the unhappy triad.

In your notebook, head the page **M2–2 The knee, anterior and from above** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain what stops the tibia sliding forward, and what happens when it fails.

4. Distinguish osteoarthritis from rheumatoid arthritis by cause, joints affected, and symmetry.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The shoulder, and why it dislocates

BRAIN DUMP FIRST

One minute. The glenohumeral joint and what actually holds it together.

Draw the humeral head against the glenoid cavity and mark how much larger the head is than the socket. Add the glenoid labrum deepening the rim, the loose capsule, and the coracohumeral and glenohumeral ligaments. Then draw the four rotator cuff muscles gripping the head, and label them.

In your notebook, head the page **M2–3 The shoulder, and why it dislocates** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say why the ligaments are not what stabilizes this joint, and name the four muscles that are.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Structure crossed with function

GRID

2 Fibrous joints

HUB

3 Synchondrosis against symphysis

SPLIT

4 A synovial joint in section

HUB

5 The six synovial types

GRID

6 The movements

GRID

7 Osteoarthritis against rheumatoid arthritis

SPLIT